Gradient-based dimension reduction

Part 2: Linear dimension reduction for regression problems

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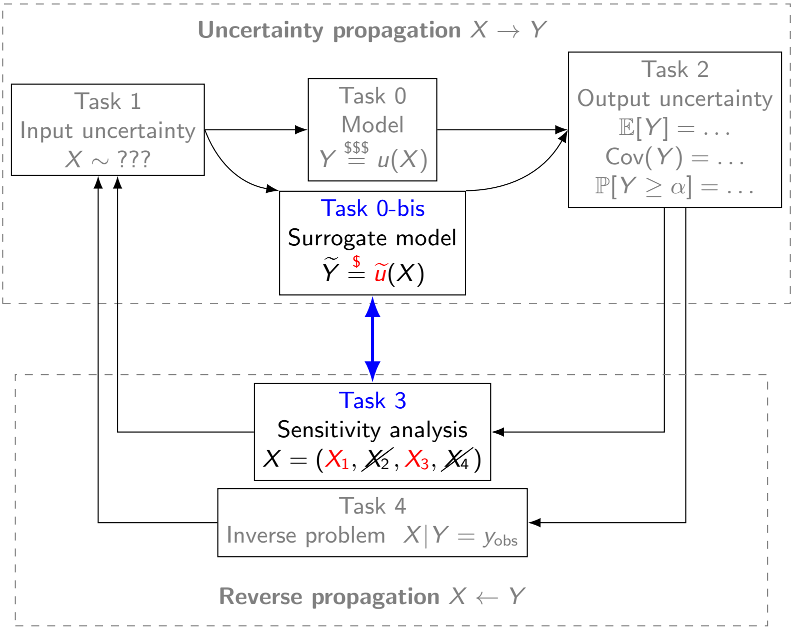
PILearnWater
Toulouse, March 2026



Uncertainty propagation $X \to Y$
Task 1
Input uncertainty
$X \sim ???$
Task 0
Model
$Y \overset{\$\$\$}{=} u(X)$
Task 2
Output uncertainty
$\mathbb{E}[Y] = \dots$
$\mathrm{Cov}(Y) = \dots$
$\mathbb{P}[Y \geq \alpha] = \dots$
Task 0-bis
Surrogate model
$\widetilde{Y} \overset{\$}{=} \widetilde{u}(X)$
Task 3
Sensitivity analysis
$X = (X_1, \cancel{X_2}, X_3, \cancel{X_4})$
Task 4
Inverse problem $X|Y = y_{\text{obs}}$
Reverse propagation $X \leftarrow Y$

Remember: **regularity is not enough!**

High-dimensional function approximation requires leveraging any **underlying low-dimensional structure** present in u, such as:

- **Sparse structure**

$$u(x) \approx \sum_{\alpha \in \Lambda_N} u_\alpha \Phi_\alpha(x), \qquad \begin{cases} \Phi_\alpha(x) : \text{ given basis} \\ \#\Lambda_N = N \end{cases}$$

- **Low-rank structure**

$$u(x) \approx \sum_{i=1}^{m} u_1^i(x_1) \dots u_d^i(x_d)$$

- **Compositional structure**

$$\boxed{u(x) \approx f \circ g(x), \qquad \begin{cases} g : \mathbb{R}^d \to \mathbb{R}^m \\ f : \mathbb{R}^m \to \mathbb{R} \\ m : \text{ latent dimension} \end{cases}}$$

- **...**

Low-effective dimension: latent dimension $m \ll d$

Given $u : \mathbb{R}^d \to \mathbb{R}$, find a **feature map** $g : \mathbb{R}^d \to \mathbb{R}^m$ and a **profile function** $f : \mathbb{R}^m \to \mathbb{R}$ with **latent dimension** $m \ll d$ such that

$$\mathbb{E}[(u(X) - f \circ g(X))^2] \leq \varepsilon$$

(regression problem in L^2)

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A trivial solution: $\{g = u \; ; \; f = id \; ; \; m = 1\}$ yields $\boxed{u = f \circ g}$. We have to restrict g in a **tractable** class

$$g \in \mathcal{G}$$

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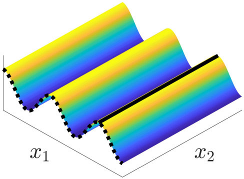

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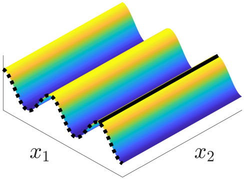

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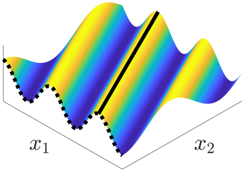

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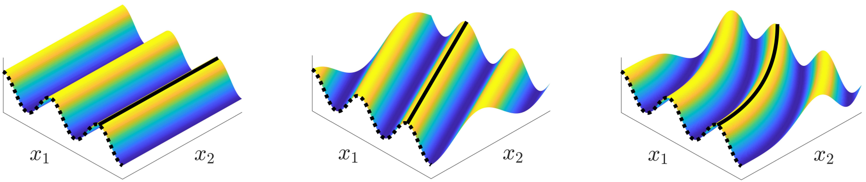

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$\mathcal{G}$ = coordinate selection
= Sensitivity Analysis

$\mathcal{G}$ = linear projection
= Active Subspace

$\mathcal{G}$ = nonlinear extraction
= Active Manifold

Part 2: Active Subspace method

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Poincaré inequality (PI)

A probability measure π_X on $\mathbb{R}^d$ satisfies the Poincaré inequality with constant $\mathbb{C}(\pi_X) < \infty$ if

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holds for any smooth function $u : \mathbb{R}^d \to \mathbb{R}$.

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If π_X satisfies

$$\overline{\mathbb{C}}(\pi_X) := \sup_{x_1} \mathbb{C}(\pi_{X_2|X_1=x_1}) < \infty$$

then there exists $f(X_1)$ (the conditional expectation $f(X_1) = \mathbb{E}[u(X)|X_1]$) such that

$$\boxed{\mathbb{E}[(u(X) - f(X_1))^2] \leq \overline{\mathbb{C}}(\pi_X)\, \mathbb{E}[(\partial_2 u(X))^2]}$$

Generalization (Active Subspace)

Given a **unitary matrix** $U = [U_m, U_\perp] \in \mathbb{R}^{d\times d}$, we decompose X as

$$X = U_m X_m + U_\perp X_\perp \qquad \begin{cases} X_m = U_m^\top X & \in \mathbb{R}^m \\ X_\perp = U_\perp^\top X & \in \mathbb{R}^{d-m} \end{cases}$$

Then, for any matrix U_m we have

$$\boxed{\begin{aligned} \min_f \mathbb{E}[(u(X) - f(U_m^\top X))^2] &\leq \overline{\mathbb{C}}(\pi_X)\mathbb{E}[\| U_\perp^\top \nabla u(X)\|^2 \\ &= \overline{\mathbb{C}}(\pi_X)\Big(\mathbb{E}[\|\nabla u(X)\|^2] - \mathbb{E}[\| U_m^\top \nabla u(X)\|^2]\Big) \\ &= \overline{\mathbb{C}}(\pi_X)\Big(\operatorname{trace}(H) - \operatorname{trace}(U_m^\top H U_m)\Big) \end{aligned}}$$

where H is the **active subspace** matrix

$$\boxed{H = \mathbb{E}[\nabla u(X) \nabla u(X)^\top]}$$

Minimizing the RHS corresponds to **compute a PCA on** $\nabla u(X)$.

$\Rightarrow$ **Active Subspace method** $\Leftarrow$

Link with **Sobol' indices**

Instead of minimizing over U_m, take $U_m = [e_i]_{i\in\tau}$ such that $\boxed{U_m^\top X = X_\tau}$ for any set of indices $\tau \subset \{1,\dots,d\}$

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where

$$H_{ii} = \mathbb{E}\left[(\partial_i u(X))^2\right]$$

are called the **Derivative based Global Sensitivity Measure** (DGSM).

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Remark: for independent inputs $X_i \perp X_j$, we even get [Zahm et.al, 2020]

$$\boxed{T_\tau \leq \sum_{i\in\tau} \frac{\mathbb{C}(\pi_{X_i})}{\mathrm{Var}(Y)} H_{ii}} \qquad \text{and} \qquad \boxed{S_\tau \geq 1 - \sum_{i\in-\tau} \frac{\mathbb{C}(\pi_{X_i})}{\mathrm{Var}(Y)} H_{ii}}$$

How to **control** $\overline{\mathbb{C}}(\pi_X)$? [Zahm et al, 2022]

Lower-bound (easy): by testing $\mathrm{Var}(f(X)) \leq \mathbb{C}(\pi_X)\mathbb{E}[\|\nabla f(x)\|^2]$ with affine functions, we get

$$\sup_{f:\text{affine}} \frac{\mathrm{Var}(f(X))}{\mathbb{E}[\|\nabla f(X)\|^2]} = \lambda_{\max}(\mathrm{Cov}(X)) \leq \mathbb{C}(\pi_X) \overset{\text{(definition of subspace PI)}}{\leq} \overline{\mathbb{C}}(\pi_X)$$

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Upper-bound (hard):

- Bakry–Émery theorem for log-concave π_X [Bakry and Émery 1983]

$$\mathrm{Hess}(-\log \pi_X(x)) \succeq \rho I_d \qquad \Rightarrow \qquad \overline{\mathbb{C}}(\pi_X) \leq 1/\rho$$

- Holley–Stroock perturbation lemma [Holley and Stroock 1987]

$$\alpha\, \rho_X(x) \leq \pi_X(x) \leq \beta\, \rho_X(x) \qquad \Rightarrow \qquad \overline{\mathbb{C}}(\pi_X) \leq \frac{\beta}{\alpha} C(\rho_X)$$

- Combination of the two above.

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Examples

- **Standard Gaussian measure** $\pi_X = \mathcal{N}(0, I_d)$: $\boxed{\overline{\mathbb{C}}(\pi_X) = 1}$ (ideal setting)
- Uniform measures on compact & convex domains
- Gaussian mixture? Yeah-ish... (worth setting)

How to **improve** $\overline{\mathbb{C}}(\pi_X)$? [Zahm et.al. 2020]

Always **whiten the input parameter**

$$\overline{X} = \mathrm{Cov}(X)^{-1/2}(X - \mathbb{E}[X])$$

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- Requires now the eigendecomposition of

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- Can be interpreted as a **supervised version** of the **unsupervised** $\mathrm{PCA}(X)$ (*i.e.* truncated Karhunen-Loeve expansion of X)

How to **further improve** $\overline{\mathbb{C}}(\pi_X)$?

[Cui et.al. 2022]: **Normalize** the parameter $\overline{X} = T(X)$ with a diffeomorphic map $T : \mathbb{R}^d \to \mathbb{R}^d$ such that $T(X) \sim \mathcal{N}(0, I_d)$ and apply the Poincaré inequality on $T(X)$:

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Alternatively: use a **Riemannian metric** for Poincaré inequalities

$$\boxed{\mathrm{Var}(u(X)) \leq C(\pi_X, W) \mathbb{E}[\|\nabla u(X)\|_{W(X)}^2]}$$

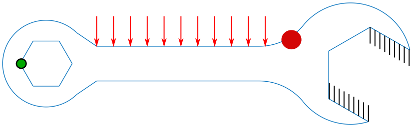
Can we define optimal metric $x \mapsto W(x)$? See [Song et al. 2019], [Heredia Joulin Roustant 2025] for product measures and [Lelièvre et al. 2025], [Cui et.al. 2024] for general measures.

Examples

A numerical example: the wrenchmark

Linear elasticity problem:

- **Input:** Young Modulus, Gaussian random field ($\overline{\mathbb{C}}(\pi_X) = 1$)
- **Output 1:** Vertical displacement of the green point
- **Output 2:** Averaged von Mises stress on the red circle



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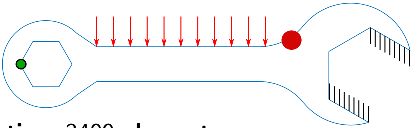

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Young modulus $= \exp(AX) \in \mathbb{R}^{3409}$,
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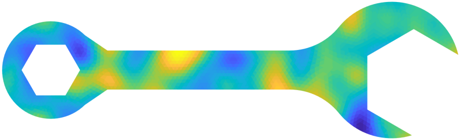

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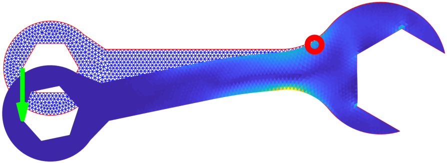

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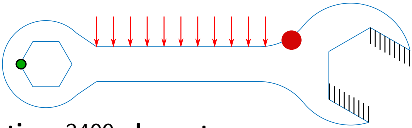

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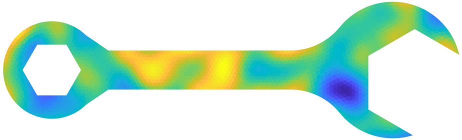

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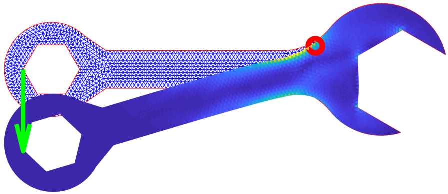

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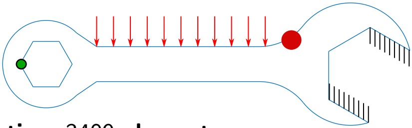

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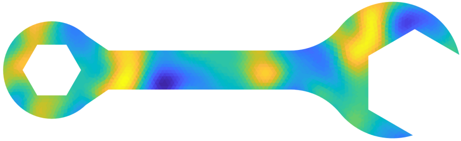
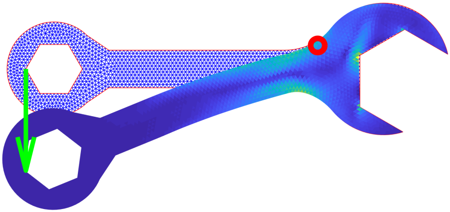
$X \sim \mathcal{N}(0, I_d)$



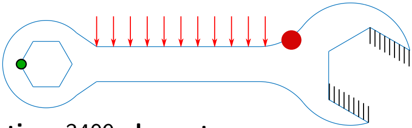
Displacement and von Mises stress



A numerical example: the wrenchmark

Linear elasticity problem:

- **Input:** Young Modulus, Gaussian random field ($\overline{\mathbb{C}}(\pi_X) = 1$)
- **Output 1:** Vertical displacement of the green point
- **Output 2:** Averaged von Mises stress on the red circle



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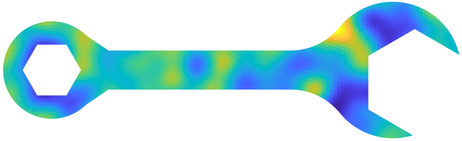

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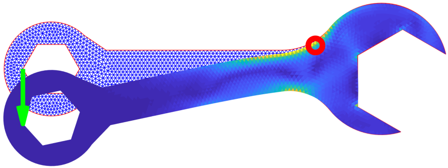
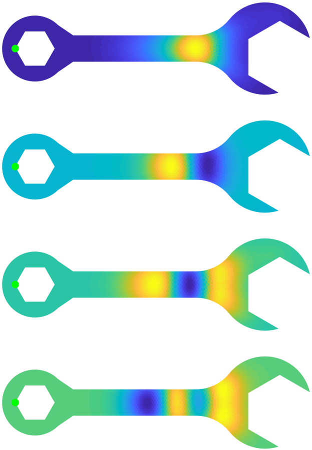


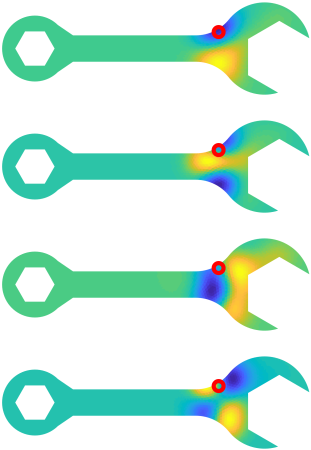
Eigenmodes $v_1, v_2, \ldots$ for $X_m = (v_1^\top X, \ldots, v_m^\top X)$

Output 1: displ. of the green point



...

Output 2: VonMises on the red circle

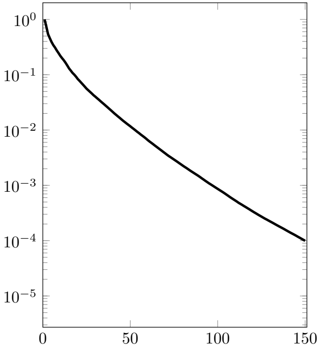


...

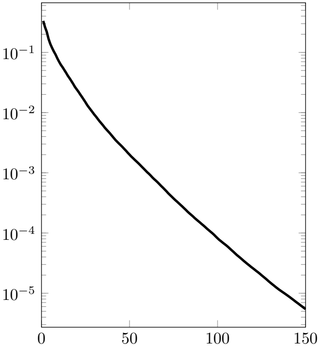
Error bound = function(m)

$$H = \mathbb{E}[\nabla u(X) \nabla u^\top(X)]$$

Output 1: displ. of the green point



Output 2: VonMises on the red circle

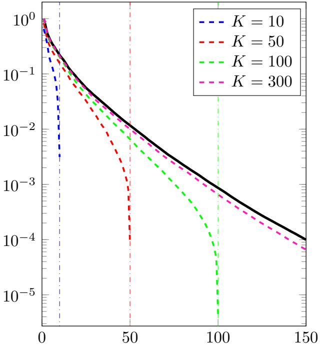


$$\{error(U_m)\} \leq \sum_{i \geq m+1} \lambda_i = function(m)$$

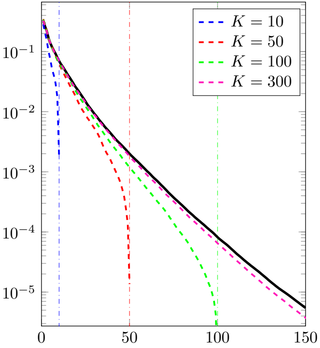
Error bound = function(m) using **MC estimation** of H

$$\hat{H} = \frac{1}{K}\sum_{i=1}^{K} \nabla u(X_i) \nabla u(X_i)^T$$

Output 1: displ. of the green point



Output 2: VonMises on the red circle



$$\{error(\hat{U}_m)\} \not\lesssim \sum_{i \geq m+1} \hat{\lambda}_i = function(m)$$ (dashed curves)

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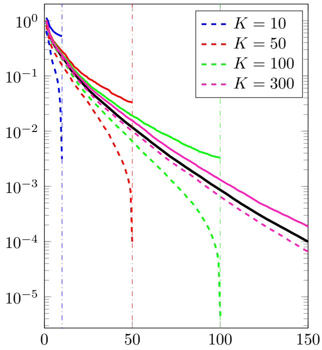
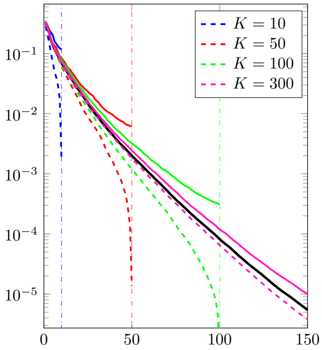
Output 1: displ. of the green point



Output 2: VonMises on the red circle



$$\{error(\hat{U}_m)\} \nleq \sum_{i\geq m+1} \hat{\lambda}_i = function(m) \qquad \text{(dashed curves)}$$

$$\{error(\hat{U}_m)\} \leq \{bound(\hat{U}_m)\} = function(m) \qquad \text{(solid curves)}$$

Linear DR **for regression**: not new, yet very active!

$$u(X_1, \ldots, X_d) \approx f(U_m^\top X)$$

Dimension reduction using the gradients of the model u,

- **[Samarov 1993]** "Exploring **regression structure** using nonparametric functional estimation", JASA
- **[Constantine et.al. 2014]** "**Active subspace** methods in theory and practice", SIAM-SISC
- **[Zahm et.al $\geq$ 2018]** Subspace Poincaré inequalities, Vector-valued models, multi-fidelity...
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or using the gradients of a surrogate model $\widetilde{u}$,

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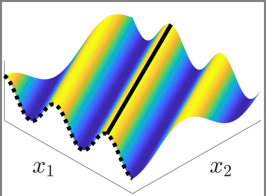
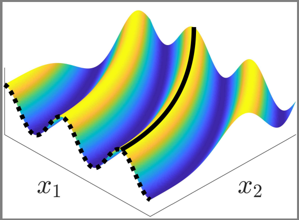
or without gradient at all!

- [Li, 1991] "**Sliced inverse regression** for dimension reduction", J. of the American Statistical Association.
- [Cha, 1994] "**Partial least squares**", Adv. Methods Mark. Res
- [Petrushev 1998] "Approximation by **ridge functions** and neural networks", SIAM-SIMA
- [Cook et.al. 2005] "**Sufficient dimension reduction** via inverse regression", JASA
- [Fukumizu, Bach, Jordan 2004] "Dimensionality reduction for supervised learning **with RKHS**", JMLR
- [....]

Towards nonlinear dimension reduction



linear



nonlinear

$$u(x) = f \circ g(x)$$

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$$\Downarrow$$

$$\nabla u(x) = \nabla g(x)^\top \nabla f(g(x))$$

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$$\Downarrow$$

$$\nabla u(x) \in \mathrm{range}(\nabla g(x)^\top)$$

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$$\nabla u(x) \in \mathrm{range}(\nabla g(x)^\top)$$

$$\Downarrow$$

$$J(g) := \mathbb{E}\left[\left\| \nabla u(X) - \underbrace{\Pi_{\mathrm{range}(\nabla g(X)^\top)}}_{\text{orthogonal projector}} \nabla u(X) \right\|^2\right] = 0$$

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Two steps strategy:

1. Build g by minimizing $g \mapsto J(g)$, (aligning ∇g with ∇u)
2. Build f by minimizing $f \mapsto \mathbb{E}[(u(X) - f \circ g(X))^2]$ (*e.g.* using KDR)

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Questions: what approximation class $\mathcal{G}$ for g in order to have

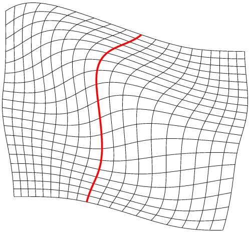
- **Q1:** the reciprocal $\Uparrow$ true,
- **Q2:** $J(g) \approx 0$ implies $u \approx f \circ g$

What approximation class $\mathcal{G}$ for g?

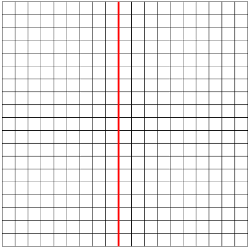
In [Verdière et.al. 2025] we employ **diffeomorphism-based** feature maps

$$\mathcal{G} \subseteq \left\{ g(x) = \begin{pmatrix} \varphi_1(x) \\ \vdots \\ \varphi_m(x) \end{pmatrix} : \quad \varphi \in \mathrm{Diffeo}(\mathbb{R}^d; \mathbb{R}^d) \right\}$$

φ is parametrized using **invertible NN**



$\xrightarrow{\varphi}$



$X \in$ Input space (IS)

$\varphi(X) \in$ Feature space (FS)

- **Q1:** Yes !
- **Q2:** Almost.................

Illustration on the **thermal block:** parametrized PDE with $d = 16$

$r = 1$						
Points	Structure	#Param	MSE%	NRMSE%	RL$_2$%	RL$_1$%
$n_{\text{train}} = 100$	BACF_IS(6)	1632	$(1.25 \pm 0.59) \times 10^{-3}$	5.18 ± 1.11	12.97 ± 2.79	8.43 ± 2.77
	BACF_IS$_{d+4}$(4)	1680	$(\mathbf{6.33} \pm 1.90) \times \mathbf{10^{-4}}$	$\mathbf{3.73} \pm 0.56$	$\mathbf{9.34} \pm 1.40$	$\mathbf{5.69} \pm 1.26$
	PFM	152	$(4.76 \pm 0.27) \times 10^{-3}$	10.34 ± 0.29	25.88 ± 0.74	20.10 ± 0.96
$n_{\text{train}} = 500$	BACF_IS(6)	1632	$(1.81 \pm 0.56) \times 10^{-4}$	2.00 ± 0.31	5.01 ± 0.77	2.19 ± 0.42
	BACF_IS$_{d+4}$(4)	1680	$(\mathbf{1.32} \pm 0.55) \times \mathbf{10^{-4}}$	$\mathbf{1.68} \pm 0.35$	$\mathbf{4.22} \pm 0.87$	$\mathbf{1.92} \pm 0.44$
	PFM	152	$(4.74 \pm 0.50) \times 10^{-3}$	10.34 ± 0.55	25.90 ± 1.39	19.79 ± 0.87
$n_{\text{train}} = 2500$	NLL	-	-	6.26	-	12.71
	DRiLLS	-	-	2.19	-	2.72

$r = 2$						
Points	Structure	#Param	MSE%	NRMSE%	RL$_2$%	RL$_1$%
$n_{\text{train}} = 100$	BACF_IS(6)	1632	$(1.27 \pm 0.37) \times 10^{-3}$	5.29 ± 0.76	13.24 ± 1.91	8.28 ± 1.75
	BACF_IS$_{d+4}$(4)	1680	$(\mathbf{1.18} \pm 0.55) \times \mathbf{10^{-3}}$	$\mathbf{5.05} \pm 1.02$	$\mathbf{12.66} \pm 2.55$	$\mathbf{7.26} \pm 1.35$
	PFM	304	$(4.65 \pm 0.27) \times 10^{-3}$	10.21 ± 0.29	25.88 ± 0.73	19.78 ± 1.00
$n_{\text{train}} = 500$	BACF_IS(6)	1632	$(2.73 \pm 4.21) \times 10^{-4}$	2.18 ± 1.17	5.47 ± 2.93	3.46 ± 3.42
	BACF_IS$_{d+4}$(4)	1680	$(\mathbf{1.42} \pm 1.20) \times \mathbf{10^{-4}}$	$\mathbf{1.70} \pm 0.55$	$\mathbf{4.25} \pm 1.38$	$\mathbf{2.39} \pm 1.73$
$n_{\text{train}} = 2500$	NLL	-	-	6.66	-	13.45
	DRiLLS	-	-	-	-	-

Table: PFM is the Polynomial Feature Maps method from **[Bigoni 2022]** and the results for DRiLLS and NLL are copied from **[Teng 2021]**.

Nonlinear feature learning: quite recent & super active!

$$u(X_1, \ldots, X_d) \approx f(g(X))$$

Nonlinear features using the gradients of the model u,

- **[Zhang et.a. 2019]** "Learning nonlinear level sets for DR in function approximation", NeurIPS
- **[Bigoni et.a. 2022]** "Nonlinear DR for surrogate modeling using gradient information", IMA:I&I.
- **[Romor et.a. 2022]** "Kernel-based active subspaces with application to computational fluid dynamics parametric problems using the discontinuous Galerkin method", Int. J. Numer. Methods Eng
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Unsupervised manifold learning

- **[Kernel PCA, Isomap, locally linear embedding (LLE), Hessian LLE, Laplacian eigenmaps, Linear discriminant analysis, Generalized discriminant analysis, (variational) autoencoders, Nonlinear Independent Component Analysis (ICA), Topological Data Analysis (TDA) ....]**

Questions?